

Huzaifa Ahmad

Metallurgical Engineer

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Professional Experience

VELOSI INTEGRITY & SAFETY PAKISTAN (Pvt) Ltd July 2023 - August 2023

Leading Inspection, Certification, Testing, Asset Integrity, HSE, Engineering & Software solutions provider

Engineer Intern

- Developed NDT proficiency in 12-inch Diameter Gas Transmission Pipelines, enhancing defect detection and ensuring product integrity using 5 NDT Techniques.
- Attained valuable insights into American Petroleum Institute (API) standards and practices (API-510, API-570)
- Mastered equipment safety and calibration for precise operations in lifting and calibration at different working sites.
- Developed technical reports for clear documentation of the Inspected Equipment of Clients.

MILLAT EQUIPMENT LIMITED

June 2022 - July 2022

Leading Manufacturing Organization in Pakistan producing High-quality Transmission gears

Engineer Intern

- Applied 5+ Heat treatment processes: Case hardening, Tempering, Aging treatment, Phosphating, Nitriding, and Induction heating in Gear Manufacturing.
- Identified 3 hazardous areas in company operations and provided solutions for them.
- Executed Carburization of 8640 Alloy steel with precision for gear manufacturing.

Education

UNIVERSITY OF THE PUNJAB

BSc. Metallurgy and Materials Engineering

Lahore, Pakistan

Oct 2020 - Aug 2024

SUPERIOR GROUP OF COLLEGES (87.91%)

FSc. Pre- Engineering

Sialkot, Pakistan

Sep 2018 - Aug 2020

Research Projects

Data-driven Prediction of Fatigue Strength in Heat-Treated 4140 Alloy Steel for Automotive Applications

Final Year Project (FYP)

- Developed **Machine Learning** models (**Random Forest, Artificial Neural Network**) to Predict fatigue strength of 4140 alloy steel.
- Analyzed the influence of key features like **chemical composition** and **heat treatment** on fatigue strength.
- **Skills:** Machine Learning (Python: Sci-kit-learn, TensorFlow), Data Analysis (Pandas, NumPy), Materials Science

Achieved high predictive accuracy for 4140 alloy steel fatigue strength with Random Forest (**R-Squared: 0.986**) and ANN (**R-Squared: 0.971**), identifying key features like chemical composition and heat treatment.

Influence of Flow Rates on Mild Steel Corrosion in Brine Solution

Corrosion Research Project

- Investigated the **Corrosion Rate** of **Mild Steel** Coupons immersed in **3.5% NaCl** solution under varying flow rates.
- Analyzed the corrosion behavior critical for industries like **Marine Engineering, Oil and Gas**, and **Infrastructure** exposed to the **Saline** environment
- **Skills:** Weight Loss Technique, Atomic Absorption Spectroscopy, SEM, XRD.

Expected Results include informing **Material Selection, Corrosion Prevention Strategies**, and **Maintenance Schedules**, aiming to enhance the durability and reliability of structures and equipment in real-life applications.

Projects

Ceramic Engineering

- Achieved a defect-free **Fritz Glaze** with a vibrant red hue and excellent gloss finish by optimizing glaze composition and application techniques.

Foundry Engineering

- Conducted research to analyze the impact of **water** percentage and **bentonite clay** concentration variations (1-6%) on the **dry compression strength** of molding sand.
- Executed **riser calculation, design**, and **3D modeling** of the casting process using Free CAD software, facilitating successful manufacturing of the given product through the casting route.

Design Engineering

- Designed and assembled a functional mechanism for converting **linear motion** to **rotational motion**, achieving a smooth simulation using **Free CAD** software

Heat Treatment

- Conducted **Normalizing** on **Plain Carbon Steel** at different **Austenitizing** temperatures, investigating Microstructure changes and Hardness variations pre- and post-treatment to analyze the effect of austenitizing temperatures.

Welding Engineering

- Designed and constructed a **Steel Truss** using bars and plates, quantified its strength through physical bend testing via welding and Universal Testing Machine (UTM), supplemented by simulation for result validation.

Technical Skills

- Materials Characterization • Materials Processing • Failure Analysis • Metallography • NDT Testing • Technical Writing
- Free CAD • MATLAB • Origin Pro • Python • Linux-Ubuntu • Machine Learning • Data Analysis

Core Competencies

- Problem Solving • Leadership • Analytical Thinking • Teamwork • Quality Control • Critical Thinking

Certifications

Google Project Management (PMP) Jan 2023 - July 2023

- Demonstrated leadership and successful project execution in achieving Google's PMP certification. ([View](#))

Creativity and Problem-Solving Skills August 2021

Metropolitan School of Business and Management (UK)

- Certified in Creativity and Problem-Solving Skills. ([View](#))

AMPP Membership Jun 2024 - May 2025

Association for Materials Performance and Protection - Member No. 11503001 ([View](#))

Young EFC Membership April 2024

Young European Federation of Corrosion

ICORR Membership Jun 2023 - Oct 2033

Institute of Corrosion – Member No. 207794

Honors and Awards

ACADEMIC SCHOLARSHIPS & AWARDS

- **1st Prize Winner** at **Pakistan Auto Show 2024** for presenting the Final Year Project organized by Pakistan Association of Automotive Parts and Accessories Manufacturers (PAAPAM) Oct 2024
- Received **Ehsaas Undergraduate Scholarship** by HEC Oct 2020 – Sep 2024
- **1st Prize Winner** at **METACOMP 1.0**, a National Metallography competition organized by UET, Lahore Feb 2022
- Received **100% Merit Scholarship** in Intermediate by Superior Group of Colleges Sep 2018 – Aug 2020
- Received **PEEF Scholarship** in Matriculation by Govt. of Punjab, Pakistan Mar 2016 – July 2018

Extracurricular Experience

SciComS-Science Communication Society Jan 2023 - Sept 2023

- **Co-Lead Team Research and Innovation:** Advanced science communication. Managed Series Posting Project & 15+ Member Team for timely, high-quality content.
- **Content Admin-LinkedIn:** Accomplished 500 followers, fastest growth, 12% CTR, demonstrating expertise, in 1 month. Refined content strategy for higher scientific community engagement.

Career Crafters April 2023 - June 2023

Ambassador

- Mentored and guided young professionals using personal career expertise through 5+ webinars.
- Supported 50+ Individuals in professional growth and goal achievement through strategic planning.